

Integrated Multiomics Reveals Alterations in Paucimannose and Complex Type *N*-Glycans in Cardiac Tissue of Patients with COVID-19

Authors

Sabarinath Peruvemba Subramanian, Melinda Wojtkiewicz, Fang Yu, Chase Castro, Erin N. Schuette, Jocelyn Rodriguez-Paar, Jared Churko, Pranav Renavikar, Daniel Anderson, Claudius Mahr, and Rebekah L. Gundry

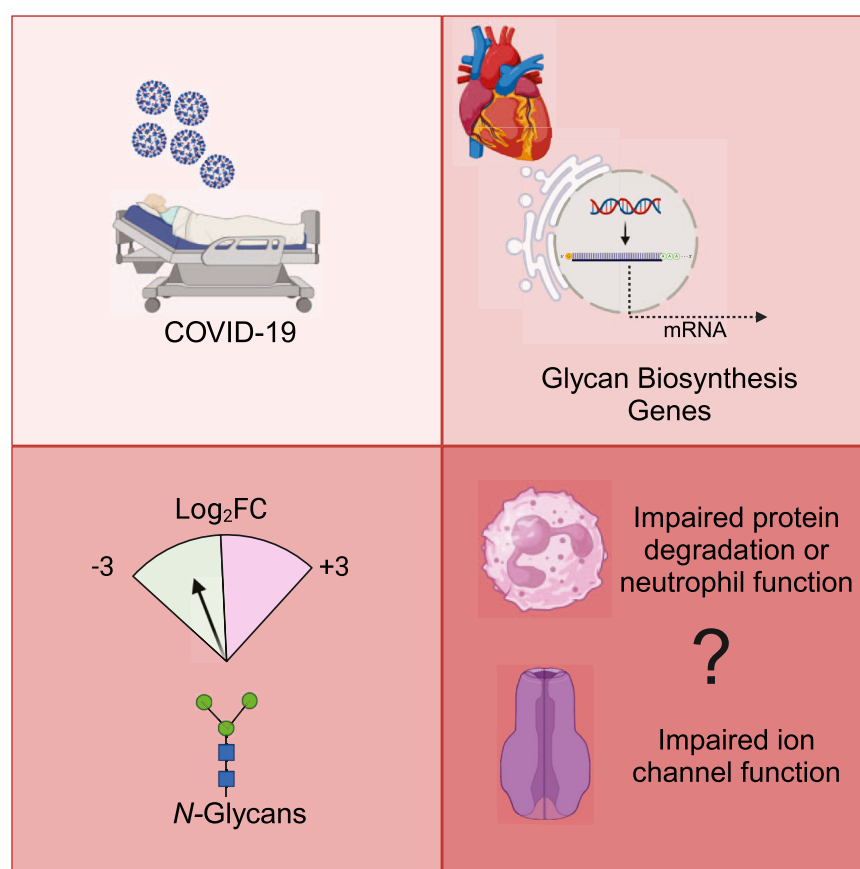
Correspondence

sabarinath.
peruvembasubramanian@zoetis.
com; Rebekah.gundry@unmc.
edu

In Brief

This study identified significant expression differences in glycozymes involved in *N*-glycan biosynthesis and a reduction in high mannose and isomers of paucimannose structures in COVID-19(+) hearts. These observations suggest that SARS-CoV-2 infection primes cardiac cells to alter the glycome with an overall trend to favor synthesis and reduce degradation. As *N*-glycosylation is critical for cardiac function, this work provides a basis for understanding the molecular events leading to cardiac damage post-COVID-19, and in the future may be extended to other viral insults and organs.

Graphical Abstract



Highlights

- Mass spectrometry and RNAseq reveal alterations in the glycome of COVID-19(+) hearts.
- COVID-19(+) hearts have altered expression of glycozymes for *N*-glycan biosynthesis.
- High mannose and paucimannose structures are decreased in COVID-19(+) hearts.

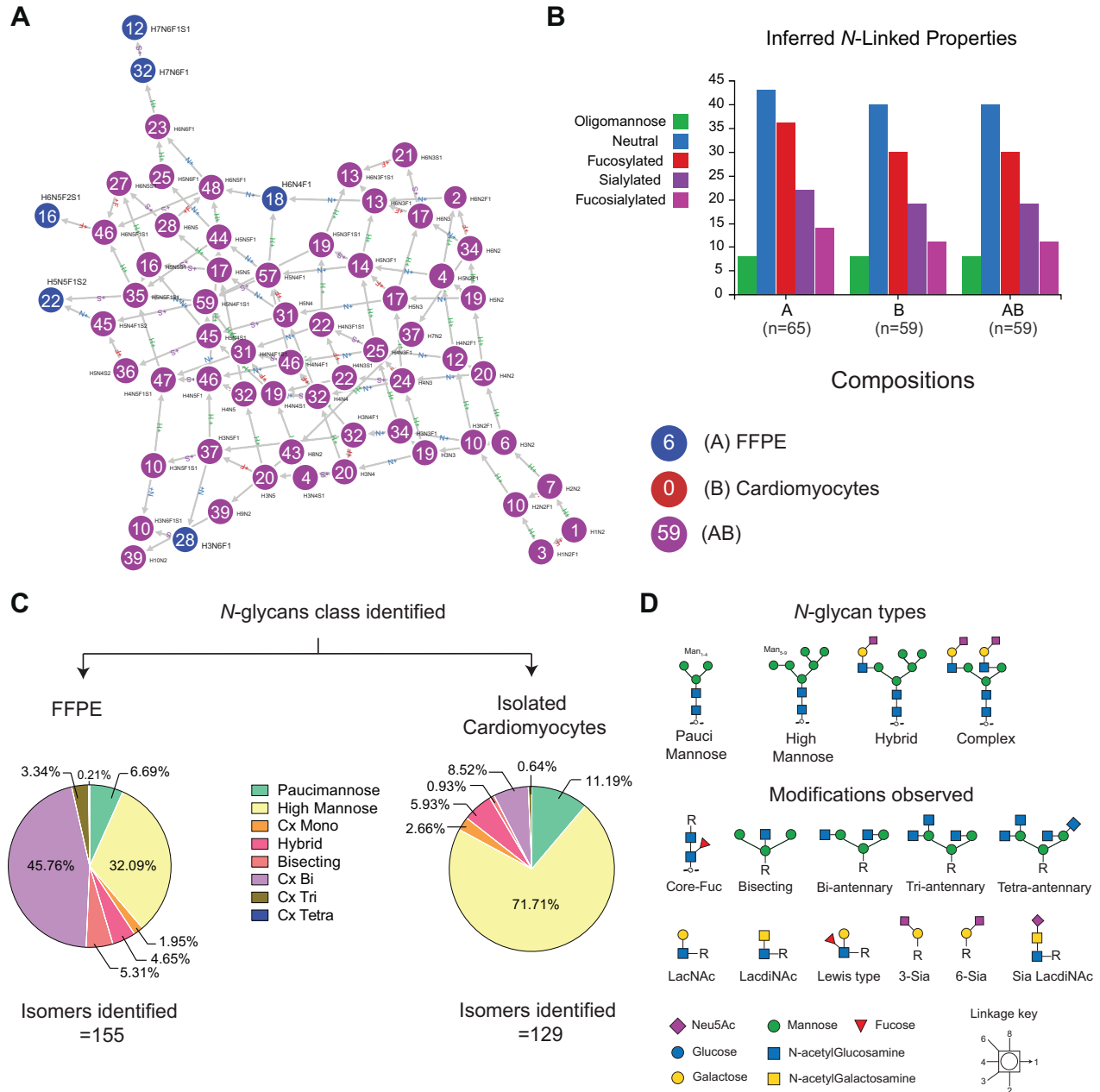
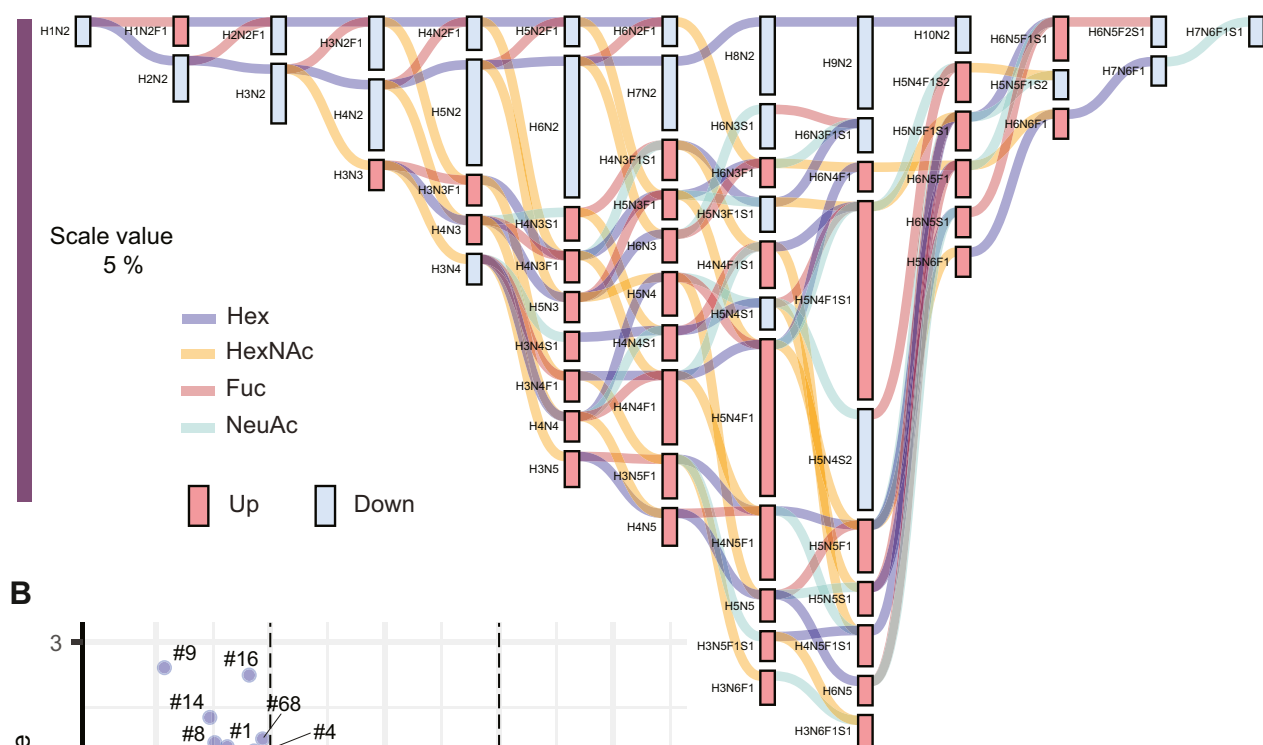


FIG. 3. Qualitative *N*-glycan profile of cardiac tissue and isolated cardiomyocytes. A, graph of relatedness of observed *N*-glycan compositions identified in FFPE cardiac tissue slices and isolated cardiomyocytes. Each node represents a unique glycan composition, and two adjacent nodes differ by one monosaccharide residue. Blue nodes are observed only in FFPE tissue, and purple is observed in both FFPE tissue and isolated cardiomyocytes. B, distribution of inferred *N*-glycan composition properties. Graphs in A and B were generated using Glyco-mpozitor (31). C, Pie charts summarizing the percentage distribution of each *N*-glycan class calculated from the relative abundance values. D, graphical representation of glycan types identified, and modifications observed at the core and terminal branches.

(~12%) *N*-glycans were significantly different between COVID-19(–) and COVID-19(+) groups (Fig. 4B). Isomers of paucimannose, high mannose, and hybrid-type glycans were significantly reduced and LacDiNAc-type structures in the complex glycan class were elevated in the COVID-19(+) group.

Unlike FFPE cardiac tissue, *N*-glycan compositions in isolated cardiomyocytes had mixed patterns with few paucimannose, high mannose, and complex structures differing in abundance (supplemental Tables S7 and S8). Specifically, *N*-glycans with compositions corresponding to paucimannoses (H1N2, H4N2, H4N2F1), high mannoses (H5N2, H5N2F1,

A Formalin Fixed Parafilm Embedded Tissue



B

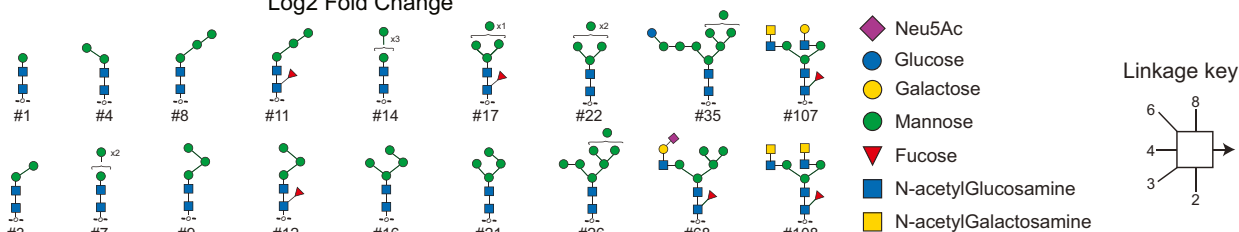
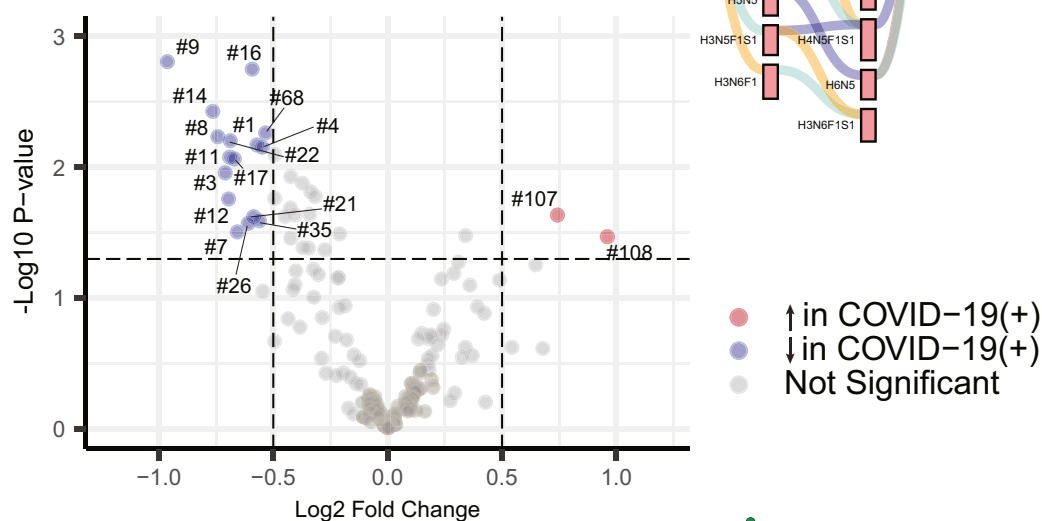


FIG. 4. Quantitative differences in the *N*-glycome of cardiac tissues (FFPE) from COVID-19(+) and COVID-19(-) patients. A, compositional analysis of *N*-glycans in cardiac tissue. Graphical results display output from differential expression analysis using Glynsight. Hex (H), HexNAc (N), Fuc (F), and Sia (S) are monosaccharide moieties that form glycan structures. Red and blue bars depict differences in relative abundance. Colored lines indicate the addition of monosaccharide moieties to a given composition. A downward trend (blue bar) in H_nN₂, corresponding to paucimannose, and oligo mannose indicates reduced levels, and an upward trend (red bars) corresponding to H_nN_nFS for complex and hybrid structures, indicates an overall increase in abundance. B, Volcano plot displaying *N*-glycan structures that are most significantly different between cardiac tissues from each patient group.

are a different patient cohort than used for glycomics; therefore, a direct correlation between glycan structures found and glycosyltransferase/hydrolase expression and their distribution in individual cell types could not be established. Second, despite differences in sample size and nuclei number, some of the differences in glycogene expression observed in the left ventricle by Delorey *et al.* (20), when compared to the right ventricle reported by Brener *et al.* (19), suggest a region-specific glycogene expression as one study used the left ventricle and the other used the right ventricle tissue. However, differences could also be due to different donor pools, inherent differences in cellular distribution, and/or the extent of infection across anatomical regions. As data comparing multiple regions within donors are not available, future analysis of age, sex, and anatomical site-matched samples of the human heart are necessary to determine if cardiac glycogene expression differences are regional. Finally, underlying comorbidities within the cohort used for our study add to the sample heterogeneity and could only be mitigated *via* statistical analysis. Given the practical challenges to biobanking during the pandemic, inclusion and exclusion criteria were not established before collection and the cohort sizes were modest, although consistent with many previously published studies of COVID-19 (19, 20).

In summary, we expanded on our prior *N*-glycome dataset for the human heart, which we expect will inform future studies designed to understand cellular and molecular changes occurring in cardiac disease. Using this library, we observed changes in *N*-glycan structure abundance that occur in the human heart post-COVID-19. Major changes in paucimannose and complex structures were observed in COVID(+) cardiac tissues. Our observations suggest that SARS-CoV-2 infection primes cardiac cells to alter the glycome at all levels, namely metabolism, transport of nucleotide sugar, and glycosyltransferases with an overall trend to favor synthesis and reduce degradation. These changes can be attributed to the viral life cycle (e.g., hijacking host machinery) or cellular transformations occurring during infection. As *N*-glycosylation is critical for maintaining electrical conductivity, cell adhesion, and protein quality control in the heart, this work provides a basis for understanding the molecular events leading to cardiac damage post-COVID-19, and in the future may be extended to other viral insults and organs.

DATA AVAILABILITY

Raw files from MS has been deposited in GlycoPOST (52). (<https://glycopost.glycosmos.org/preview/1409704126670147788a822>. Pin=7380).

Supplemental data—This article contains [supplemental data](#) (53).

Acknowledgments—Shelby Konfrst, Kyu Shik Kim, and Kimberley Sinanan extracted clinical data from patient medical records. Isaac Blakeslee assisted with snRNA-seq data analysis. Maria Burkovetskaya performed initial testing of lectins for imaging. Aspects of this work were, in part, inspired by discussions at the Dagstuhl Seminar 23301: Computational Proteomics held in July 2023 at Schloss Dagstuhl, Wadern, Germany.

Funding and additional information—This work was supported by the American Heart Association Rapid Response Grant, and National Institutes of Health (R33HL154123 and R35HL155460) to R. G. Funding sources were not involved in study design, data collection, interpretation, analysis or publication. MS analyses were performed using instrumentation in the CardiOmics Program in the Center for Heart and Vascular Research at UNMC. Tissues for histological and glycomics analysis were provided by the Nebraska Cardiovascular Biobank and Registry in the Center for Heart and Vascular Research at UNMC, in collaboration with Dr Stanley Radio, Department Pathology and Microbiology, UNMC. Tissue sectioning was performed by the Tissue Science Facility core lab at UNMC.

Author contributions—C. C., F. Y., M. W., S. P. S., P. R., J. R., E. N. S., D. R. A., R. L. G., and C. M. writing—original draft; C. C., M. W., S. P. S., E. N. S., and R. L. G. visualization; C. C., F. Y., M. W., S. P. S., J. C., J. R., E. N. S., D. R. A., and R. L. G. methodology; C. C., F. Y., M. W., S. P. S., P. R., J. R., E. N. S., D. R. A., R. L. G., and C. M. investigation; C. C., F. Y., M. W., S. P. S., P. R., J. C., J. R., E. N. S., D. R. A., R. L. G., and C. M. formal analysis; F. Y., M. W., S. P. S., P. R., J. C., J. R., E. N. S., D. R. A., R. L. G., and C. M. writing—review & editing; M. W., S. P. S., and E. N. S. data curation; M. W., S. P. S., and R. L. G. conceptualization; S. P. S. and R. L. G. project administration; R. L. G. supervision; R. L. G. resources, R. L. G. funding acquisition.

Conflict of interest—R. L. G. is on the advisory board of ProtiFi, LLC, and receives no compensation of any kind.

Abbreviations—The abbreviations used are: AAL, aleuria aurantia lectin; COVID-19, Coronavirus infectious disease of 2019; ECL, erythrina cristagalli lectin; FDR, false discovery rate; FFPE, formalin fixed paraffin embedded; Fuc (F), fucose; H and E, hematoxylin and eosin; Hex (H), hexose; hexnac (N), N-acetyl hexosamine; MAL, maackia amurensis lectin; mnesi, multinozzle electrospray ionization; MS, mass spectrometry; PGC, porous graphitized column; pngase F, peptide N-glycanase F; PNA, peanut agglutinin; RCA, ricinus communis agglutinin; SARS-cov-2, severe acute respiratory syndrome coronavirus 2; Sia (S), sialic acid; snRNA-seq, single nuclei RNA sequencing; SNA, sambucus nigra agglutinin; VVA, vicia villosa lectin.

Received October 7, 2024, and in revised form, February 19, 2025
Published, MCPRO Papers in Press, February 22, 2025, <https://doi.org/10.1016/j.mcpro.2025.100929>

REFERENCES

- Zhu, N., Zhang, D., Wang, W., Li, X., Yang, B., Song, J., *et al.* (2020) A novel coronavirus from patients with pneumonia in China, 2019. *N. Engl. J. Med.* **382**, 727–733
- Goerlich, E., Chung, T. H., Hong, G. H., Metkus, T. S., Gilotra, N. A., Post, W. S., *et al.* (2024) Cardiovascular effects of the post-COVID-19 condition. *Nat. Cardiovasc. Res.* **3**, 118–129
- Xie, Y., Xu, E., Bowe, B., and Al-Aly, Z. (2022) Long-term cardiovascular outcomes of COVID-19. *Nat. Med.* **28**, 583–590
- Nishiga, M., Wang, D. W., Han, Y., Lewis, D. B., and Wu, J. C. (2020) COVID-19 and cardiovascular disease: from basic mechanisms to clinical perspectives. *Nat. Rev. Cardiol.* **17**, 543–558
- Adeghate, E. A., Eid, N., and Singh, J. (2021) Mechanisms of COVID-19-induced heart failure: a short review. *Heart Fail. Rev.* **26**, 363–369
- Conway, E. M., Mackman, N., Warren, R. Q., Wolberg, A. S., Mosnier, L. O., Campbell, R. A., *et al.* (2022) Understanding COVID-19-associated coagulopathy. *Nat. Rev. Immunol.* **22**, 639–649
- Perico, L., Benigni, A., Casiraghi, F., Ng, L. F. P., Renia, L., and Remuzzi, G. (2021) Immunity, endothelial injury and complement-induced coagulopathy in COVID-19. *Nat. Rev. Nephrol.* **17**, 46–64
- Noris, M., Benigni, A., and Remuzzi, G. (2020) The case of complement activation in COVID-19 multiorgan impact. *Kidney Int.* **98**, 314–322
- Akhmerov, A., and Marbán, E. (2020) COVID-19 and the heart. *Circ. Res.* **126**, 1443–1455
- Watanabe, Y., Bowden, T. A., Wilson, I. A., and Crispin, M. (2019) Exploitation of glycosylation in enveloped virus pathobiology. *Biochim. Biophys. Acta Gen. Subj.* **1863**, 1480–1497
- Aloor, A., Aradhya, R., Venugopal, P., Gopalakrishnan Nair, B., and Suravajhala, R. (2022) Glycosylation in SARS-CoV-2 variants: a path to infection and recovery. *Biochem. Pharmacol.* **206**, 115335
- Pongracz, T., Vidarsson, G., and Wuhrer, M. (2022) Antibody glycosylation in COVID-19. *Glycoconj. J.* **39**, 335–344
- Qin, R., Kurz, E., Chen, S., Zeck, B., Chiribogás, L., Jackson, D., *et al.* (2022) α 2,6-Sialylation is upregulated in severe COVID-19, implicating the complement cascade. *ACS Infect. Dis.* **8**, 2348–2361
- Zhang, L., Mann, M., Syed, Z. A., Reynolds, H. M., Tian, E., Samara, N. L., *et al.* (2021) Furin cleavage of the SARS-CoV-2 spike is modulated by O-glycosylation. *Proc. Natl. Acad. Sci. U. S. A.* **118**, e2109905118
- Tétrault, M.-P., Bourdin, B., Briot, J., Segura, E., Lesage, S., Fiset, C., *et al.* (2016) Identification of glycosylation sites essential for surface expression of the CaV α 2 δ 1 subunit and modulation of the cardiac CaV1.2 channel activity. *J. Biol. Chem.* **291**, 4826–4843
- Goth, C. K., Petäjä-Repo, U. E., and Rosenkilde, M. M. (2020) G protein-coupled receptors in the sweet spot: glycosylation and other post-translational modifications. *ACS Pharmacol. Transl. Sci.* **3**, 237–245
- Ednie, A. R., Deng, W., Yip, K.-P., and Bennett, E. S. (2019) Reduced myocyte complex N-glycosylation causes dilated cardiomyopathy. *FASEB J.* **33**, 1248–1261
- Ednie, A. R., Parrish, A. R., Sonner, M. J., and Bennett, E. S. (2019) Reduced hybrid/complex N-glycosylation disrupts cardiac electrical signaling and calcium handling in a model of dilated cardiomyopathy. *J. Mol. Cell Cardiol.* **132**, 13–23
- Brener, M. I., Hulke, M. L., Fukuma, N., Golob, S., Zilinyi, R. S., Zhou, Z., *et al.* (2022) Clinico-histopathologic and single-nuclei RNA-sequencing insights into cardiac injury and microthrombi in critical COVID-19. *JCI Insight* **7**. <https://doi.org/10.1172/jci.insight.154633>
- Delorey, T. M., Ziegler, C. G. K., Heimberg, G., Normand, R., Yang, Y., Segerstolpe, A., *et al.* (2021) COVID-19 tissue atlases reveal SARS-CoV-2 pathology and cellular targets. *Nature* **595**, 107–111
- Huang, Y.-F., Aoki, K., Akase, S., Ishihara, M., Liu, Y.-S., Yang, G., *et al.* (2021) Global mapping of glycosylation pathways in human-derived cells. *Dev. Cell* **56**, 1195–1209.e7
- Berg Luecke, L., Waas, M., Littrell, J., Wojtkiewicz, M., Castro, C., Burkovetskaya, M., *et al.* (2023) Surfaceome mapping of primary human heart cells with CellSurfer uncovers cardiomyocyte surface protein LSMEM2 and proteome dynamics in failing hearts. *Nat. Cardiovasc. Res.* **2**, 76–95
- Wojtkiewicz, M., Luecke, L. B., Castro, C., Burkovetskaya, M., Mesidor, R., and Gundry, R. L. (2021) Bottom-up proteomic analysis of human adult cardiac tissue and isolated cardiomyocytes. *J. Mol. Cell Cardiol.* <https://doi.org/10.1016/j.jmcc.2021.08.008>
- Cardiff, R. D., Miller, C. H., and Munn, R. J. (2014) Manual hematoxylin and eosin staining of mouse tissue sections. *Cold Spring Harb. Protoc.* **2014**, 655–658
- Wojtkiewicz, M., Subramanian, S. P., and Gundry, R. L. (2024) Multinozzle emitter for improved negative mode analysis of reduced native N-glycans by microflow porous graphitized carbon liquid chromatography mass spectrometry. *Anal. Chem.* **96**, 5746–5751
- Adams, K. J., Pratt, B., Bose, N., Dubois, L. G., St John-Williams, L., Perrott, K. M., *et al.* (2020) Alzheimer’s disease metabolomics consortium, skyline for small molecules: a unifying software package for quantitative metabolomics. *J. Proteome Res.* **19**, 1447–1458
- Ceroni, A., Maass, K., Geyer, H., Geyer, R., Dell, A., and Haslam, S. M. (2008) GlycoWorkbench: a tool for the computer-assisted annotation of mass spectra of glycans. *J. Proteome Res.* **7**, 1650–1659
- Abrahams, J. L., Campbell, M. P., and Packer, N. H. (2018) Building a PGC-LC-MS N-glycan retention library and elution mapping resource. *Glycoconj. J.* **35**, 15–29
- Domon, B., and Costello, C. E. (1988) A systematic nomenclature for carbohydrate fragmentations in FAB-MS/MS spectra of glycoconjugates. *Glycoconj. J.* **5**, 397–409
- Harvey, D. J. (2020) Negative ion mass spectrometry for the analysis of N-linked glycans. *Mass Spectrom. Rev.* **39**, 586–679
- Robin, T., Mariethoz, J., and Lisacek, F. (2020) Examining and fine-tuning the selection of glycan compositions with GlyConnect compozitor. *Mol. Cell Proteomics* **19**, 1602–1618
- Alloci, D., Ghraichy, M., Barletta, E., Gastaldello, A., Mariethoz, J., and Lisacek, F. (2018) Understanding the glycome: an interactive view of glycosylation from glycompositions to glycoepitopes. *Glycobiology* **28**, 349–362
- Subramanian, S. P., and Gundry, R. L. (2024) Integration of web-based tools to visualize, integrate, and interpret glycogene expression and glycomics data. *Methods Mol. Biol.* **2836**, 97–109
- York, W. S., Mazumder, R., Ranzinger, R., Edwards, N., Kahsay, R., Aoki-Kinoshita, K. F., *et al.* (2019) GlyGen: computational and informatics Resources for glycoscience. *Glycobiology*. <https://doi.org/10.1093/glycob/cwz080>
- Basso, C., Leone, O., Rizzo, S., De Gaspari, M., van der Wal, A. C., Aubry, M.-C., *et al.* (2020) Pathological features of COVID-19-associated myocardial injury: a multicentre cardiovascular pathology study. *Eur. Heart J.* **41**, 3827–3835
- Dal Ferro, M., Bussani, R., Paldino, A., Nuzzi, V., Collesi, C., Zentilin, L., *et al.* (2021) SARS-CoV-2, myocardial injury and inflammation: insights from a large clinical and autopsy study. *Clin. Res. Cardiol.* **110**, 1822–1831
- Ashwood, C., Waas, M., Weerasekera, R., and Gundry, R. L. (2020) Reference glycan structure libraries of primary human cardiomyocytes and pluripotent stem cell-derived cardiomyocytes reveal cell-type and culture stage-specific glycan phenotypes. *J. Mol. Cell Cardiol.* **139**, 33–46
- Kanamaru, K., Cranley, J., Muraro, D., Miranda, A. M. A., Ho, S. Y., Wilbrey-Clark, A., *et al.* (2023) Spatially resolved multiomics of human cardiac niches. *Nature* **619**, 801–810
- Ghosh, S., Dellibovi-Ragheb, T. A., Kerviel, A., Pak, E., Qiu, Q., Fisher, M., *et al.* (2020) β -Coronaviruses use lysosomes for egress instead of the biosynthetic secretory pathway. *Cell* **183**, 1520–1535.e14
- Winchester, B. (2005) Lysosomal metabolism of glycoproteins. *Glycobiology* **15**, 1R–15R
- Altmann, F., Schwihla, H., Staudacher, E., Glössl, J., and März, L. (1995) Insect cells contain an unusual, membrane-bound beta-N-acetylglucosaminidase probably involved in the processing of protein N-glycans. *J. Biol. Chem.* **270**, 17344–17349
- Thaysen-Andersen, M., Venkatakrishnan, V., Loke, I., Laurini, C., Diestel, S., Parker, B. L., *et al.* (2015) Human neutrophils secrete bioactive paucimannosidic proteins from azurophilic granules into pathogen-infected sputum. *J. Biol. Chem.* **290**, 8789–8802
- Jinnelov, A., Ali, L., Tinti, M., Güther, M. L. S., and Ferguson, M. A. J. (2017) Single-subunit oligosaccharyltransferases of *Trypanosoma brucei* display

- different and predictable peptide acceptor specificities. *J. Biol. Chem.* **292**, 20328–20341
44. He, W., Gao, Y., Zhou, J., Shi, Y., Xia, D., and Shen, H.-M. (2022) Friend or Foe? Implication of the autophagy-lysosome pathway in SARS-CoV-2 infection and COVID-19. *Int. J. Biol. Sci.* **18**, 4690–4703
 45. Yang, N., and Shen, H.-M. (2020) Targeting the endocytic pathway and autophagy process as a novel therapeutic strategy in COVID-19. *Int. J. Biol. Sci.* **16**, 1724–1731
 46. Mijaljica, D., and Klionsky, D. J. (2020) Autophagy/virophagy: a “disposal strategy” to combat COVID-19. *Autophagy* **16**, 2271–2272
 47. Varki, A., Cummings, R. D., Esko, J. D., Stanley, P., Hart, G. W., Aebi, M., *et al.*, eds. (2022) *Essentials of Glycobiology*, 4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor (NY)
 48. Loke, I., Østergaard, O., Heegaard, N. H. H., Packer, N. H., and Thaysen-Andersen, M. (2017) Paucimannose-rich N-glycosylation of spatiotemporally regulated human neutrophil elastase modulates its immune functions. *Mol. Cell Proteomics* **16**, 1507–1527
 49. Dahmen, A.-C., Fergen, M.-T., Laurini, C., Schmitz, B., Loke, I., Thaysen-Andersen, M., *et al.* (2015) Paucimannosidic glycoepitopes are functionally involved in proliferation of neural progenitor cells in the subventricular zone. *Glycobiology* **25**, 869–880
 50. Chen, M., Assis, D. M., Benet, M., McClung, C. M., Gordon, E. A., Ghose, S., *et al.* (2023) Comparative site-specific N-glycoproteome analysis reveals aberrant N-glycosylation and gives insights into mannose-6-phosphate pathway in cancer. *Commun. Biol.* **6**, 1–17
 51. Guillard, M., Dimopoulou, A., Fischer, B., Morava, E., Lefeber, D. J., Kornak, U., *et al.* (2009) Vacuolar H⁺-ATPase meets glycosylation in patients with cutis laxa. *Biochim. Biophys. Acta* **1792**, 903–914
 52. Watanabe, Y., Aoki-Kinoshita, K. F., Ishihama, Y., and Okuda, S. (2021) GlycoPOST realizes FAIR principles for glycomics mass spectrometry data. *Nucleic Acids Res.* **49**, D1523–D1528
 53. Liew, C. Y., Chen, J.-L., Lin, Y.-T., Luo, H.-S., Hung, A.-T., Magoling, B. J. A., *et al.* (2024) Chromatograms and mass spectra of high-mannose and paucimannose N-glycans for rapid isomeric identifications. *J. Proteome Res.* **23**, 939–955